

Introduction

We consider the problem of predicting the acoustic field generated by one or more HVAC units located outdoors in a residential neighborhood. The receivers of interest are individual houses, modeled as points in space at a specified height above the ground. The propagation environment includes free-field spreading, ground reflection, and diffraction by intervening houses modeled as knife-edge obstacles.

The purpose of this document is to construct a complete, self-contained, frequency-domain derivation of the complex acoustic pressure field at each receiver. All propagation physics are performed in the linear complex-pressure domain. Decibels are used only for input source levels and final output levels. The final result is a unified expression for the C-weighted sound pressure level at a receiver for any of three HVAC configurations:

- a single point source,
- a line of HVAC units,
- a square array of HVAC units.

Assumptions

The propagation model developed in this document is formulated in a 2D vertical plane and is intended to capture the dominant physical mechanisms governing sound transmission from residential HVAC equipment to nearby receivers. The following assumptions define the scope and limitations of the model:

- **Two-dimensional diffraction geometry.** Each obstacle is idealized as an infinite straight knife-edge, which reduces the diffraction calculation to a 2D vertical cross-section per source–receiver path. Source and receiver locations remain point locations in 3D space; propagation along the direct, reflected, and diffracted paths uses full 3D spherical spreading. The “2D” label refers to the edge geometry, not the propagation itself.
- **Primary-path diffraction only.** Diffraction is modeled exclusively along the direct source–receiver path. Off-axis diffraction, lateral scattering, and contributions from secondary propagation paths are neglected as higher-order effects.
- **Obstacles represented as knife edges.** Each house is modeled as a single effective knife edge located at the roof ridge or leading roof edge. Diffraction from roof slopes, pitched surfaces, and other smooth geometrical features is not included.
- **Shared edge across multiple sources.** For the line and square-array configurations, all sources are assumed to resolve to the same effective edge point on a given obstacle – i.e., the roofline is treated as effectively infinite/straight relative to the lateral extent of the source array. This assumption breaks down if the array’s lateral extent is not small compared to the Fresnel zone width at the edge: sources near the ends of the array may see a different diffraction geometry, or may diffract around the end of a finite-length roofline rather than over it. This is a particular risk for the square-array configuration, since it spans two horizontal dimensions against what is fundamentally a line-like obstacle. Users of this model should verify that array extent versus edge length and versus Fresnel-zone width remains within the regime where a single shared edge point is a reasonable approximation.

- **No scattering from roof planes.** Roof surfaces are treated as smooth, non-diffracting elements. Only the geometric discontinuity at the roof edge contributes to diffraction.
- **Homogeneous ground.** The ground is modeled as a homogeneous surface (e.g., grass or farmland) using an image-source formulation. Spatial variations such as roads, driveways, or mixed-impedance patches are not included.
- **No meteorological effects.** Atmospheric refraction, wind gradients, temperature gradients, and other meteorological influences are not modeled.
- **Monopole source representation.** HVAC units are modeled as monopole or equivalent point sources. Directivity effects are not included unless explicitly added.
- **Linear acoustics.** All propagation is assumed linear. Nonlinear effects and turbulence are not considered.

1 Derivation of the complex acoustic field

We now derive the full expression for the complex acoustic pressure field at a receiver, beginning with the Helmholtz equation and proceeding through propagation, ground reflection, diffraction, and source configuration.

1. Free-field Green's function

Let the acoustic pressure field $p(\mathbf{r}, \omega)$ satisfy the Helmholtz equation

$$(\nabla^2 + k_{air}^2(\omega)) p(\mathbf{r}, \omega) = -S(\omega) \delta(\mathbf{r} - \mathbf{r}_S),$$

where

$$k_{air}(\omega) = \frac{\omega}{c_{air}(\omega)} + i \alpha_{air}(\omega)$$

is the complex wave number.

The free-field Green's function is

$$G(\mathbf{r}, \mathbf{r}_S, \omega) = \frac{e^{-ik_{air}(\omega)\|\mathbf{r}-\mathbf{r}_S\|}}{\|\mathbf{r} - \mathbf{r}_S\|}.$$

Thus the free-field pressure from a single source is

$$p_{free}(\mathbf{r}, \omega) = S(\omega) G(\mathbf{r}, \mathbf{r}_S, \omega).$$

2. Ground reflection via image source

Let the ground be the plane $z = 0$. The image source is

$$\mathbf{r}'_S = (x_S, y_S, -h_s).$$

Let $R_g(\omega)$ be the complex ground reflection coefficient. The total pressure including the ground reflection is

$$p_{prop}(\mathbf{r}, \omega) = S(\omega) \left[\frac{e^{-ik_{air}(\omega)R_d}}{R_d} + R_g(\omega) \frac{e^{-ik_{air}(\omega)R_r}}{R_r} \right],$$

where

$$R_d = \|\mathbf{r} - \mathbf{r}_S\|, \quad R_r = \|\mathbf{r} - \mathbf{r}'_S\|.$$

3. Multi-edge diffraction via complex Fresnel integrals

Let there be N diffracting edges between the source region and the receiver. For edge i , define the Fresnel parameter

$$\nu_i(\omega) = h_i \sqrt{\frac{2}{\lambda(\omega)} \left(\frac{1}{d_{si}} + \frac{1}{d_{ir}} \right)}, \quad \lambda(\omega) = \frac{2\pi c_{air}(\omega)}{\omega}.$$

Define the Fresnel integrals using the standard normalized convention

$$C(\nu) = \int_0^\nu \cos\left(\frac{\pi t^2}{2}\right) dt, \quad S(\nu) = \int_0^\nu \sin\left(\frac{\pi t^2}{2}\right) dt,$$

so that $C(\nu), S(\nu) \rightarrow \pm \frac{1}{2}$ as $\nu \rightarrow \pm\infty$.

The complex diffraction factor for edge i is

$$D_i(\omega) = \frac{1+i}{2} \left[\left(\frac{1}{2} - C(\nu_i(\omega)) \right) - i \left(\frac{1}{2} - S(\nu_i(\omega)) \right) \right].$$

This form is fixed relative to an earlier draft, which omitted the leading $\frac{1+i}{2}$ normalization factor. Without it, $|D_i|$ does not converge to the correct limiting values: the corrected formula gives $D_i \rightarrow 1$ (no loss) in the fully illuminated limit $\nu_i \rightarrow -\infty$, $|D_i| = \frac{1}{2}$ (−6 dB) exactly at the shadow boundary $\nu_i = 0$, and $D_i \rightarrow 0$ deep in shadow ($\nu_i \rightarrow +\infty$) – consistent with standard Fresnel/Maekawa knife-edge diffraction behavior. The integral convention was simultaneously corrected to include the $\pi/2$ factor in the integrand; without it, C and S asymptote to $\sqrt{\pi/8} \approx 0.627$ rather than $\frac{1}{2}$, which broke the limiting behavior of D_i even with the normalization factor restored.

The total diffraction factor is

$$D_{tot}(\omega) = \prod_{i=1}^N D_i(\omega).$$

Thus the diffracted field from a single source is

$$p_{diff}(\mathbf{r}, \omega) = D_{tot}(\omega) p_{prop}(\mathbf{r}, \omega).$$

4. Multiple HVAC units and interference

Let the HVAC configuration consist of M sources at positions $\mathbf{r}_j$ with spectra $S_j(\omega)$. For each source j , define

$$p_{diff,j}(\mathbf{r}, \omega) = D_{tot}(\omega) S_j(\omega) \left[\frac{e^{-ik_{air}(\omega)R_{d,j}}}{R_{d,j}} + R_g(\omega) \frac{e^{-ik_{air}(\omega)R_{r,j}}}{R_{r,j}} \right].$$

By linearity of the Helmholtz equation, the total field is

$$p_{tot}(\mathbf{r}, \omega) = \sum_{j=1}^M p_{diff,j}(\mathbf{r}, \omega).$$

This expression automatically includes full complex interference between all sources.

5. HVAC configurations

Point source: $M = 1$.

Line of sources:

$$\mathbf{r}_j = (x_s, y_j, h_s), \quad y_j = \left(j - \frac{N+1}{2}\right) \Delta y.$$

Square array:

$$\mathbf{r}_{m,n} = \left(x_s + \left(m - \frac{N_x+1}{2}\right) \Delta x, y_s + \left(n - \frac{N_y+1}{2}\right) \Delta y, h_s\right).$$

6. Conversion to SPL and C-weighting

The unweighted spectral level is

$$L_p(\mathbf{r}, \omega) = 20 \log_{10} \left(\frac{|p_{tot}(\mathbf{r}, \omega)|}{p_{ref}} \right).$$

The C-weighted level is

$$L_C(\mathbf{r}, \omega) = L_p(\mathbf{r}, \omega) + C_w(\omega).$$

Next Steps: Validation

Although the propagation and diffraction model developed in this document is derived directly from standard wave-based acoustics and the complex Fresnel formulation, full validation against external benchmark datasets remains a recommended next step. The following reference geometry and validation procedure outline how such checks should be performed.

Reference geometry for validation

A representative two-house configuration is proposed for testing the multi-edge diffraction behavior. The HVAC source is placed at (0, 2 m). A one-story house with roof edge at (15 m, 4 m) lies between the source and a two-story house with roof edge at (20 m, 7 m). The receiver is positioned behind the two-story house at (25 m, 1.8 m), corresponding to a listener height of approximately six feet. In this geometry, both roof edges lie above the source–receiver line of sight, producing a double-shadow region that serves as a stringent test of the multi-edge diffraction formulation.

Recommended validation procedure

To confirm correct implementation of the model, the following three-stage validation plan is recommended:

1. **Single-edge Fresnel diffraction.** Compare the predicted single-edge diffraction factor with canonical Fresnel benchmark curves. Agreement verifies the correctness of the Fresnel parameter $\nu(\omega)$ and the integrals $C(\nu)$ and $S(\nu)$.
2. **Double-edge diffraction.** Compare the sequential product $D_1(\omega)D_2(\omega)$ with published two-knife-edge benchmark cases. This confirms that the multi-edge diffraction behavior is implemented correctly, especially when edges are closely spaced.
3. **Outdoor building and barrier data.** Compare the overall attenuation for the reference geometry with typical measured losses reported for building-edge and barrier diffraction in outdoor acoustics literature. This provides a practical check that the predicted attenuation falls within the expected physical range for residential-scale obstacles.

Limitations and scope

The validation plan above relies on component-level benchmark datasets rather than a single measurement matching the exact two-house geometry. No public dataset exists that reproduces this specific configuration. The following modeling assumptions also apply:

- Diffraction is modeled only along the primary source–receiver path. Off-axis diffraction, lateral scattering, and diffraction from roof slopes are neglected as higher-order effects. Each house is treated as a single effective knife edge.
- Ground reflection is modeled using an image-source formulation with homogeneous ground impedance. Spatial variations in ground type (e.g., roads, driveways) are not included.
- The validation steps described above are intended as future work to confirm numerical implementation, not as results already obtained.

Appendix A: Typical Sound Pressure Levels

Level (dB)	Example
30	Quiet bedroom at night
40	Quiet library
50	Light rainfall
60	Normal conversation
70	Household vacuum cleaner
80	Busy street traffic
90	Gas lawnmower
100	Jackhammer
110	Rock concert
120	Threshold of discomfort

Appendix B: Numerical Behavior Checklist

This appendix provides a set of numerical cues and qualitative expectations that can be used to verify correct implementation of the propagation and diffraction model. These checks are intended as practical guidance during development and debugging. They do not constitute validation against external datasets; rather, they ensure that each component of the model behaves consistently with known analytical properties of wave-based acoustics.

B.1 Single-edge diffraction

For a single knife-edge placed between the source and receiver:

- The Fresnel parameter $\nu(\omega)$ should increase monotonically with frequency for fixed geometry.
- The magnitude of the diffraction factor $|D(\omega)|$ should decrease as ν increases, corresponding to deeper shadowing at higher frequencies.
- In the fully illuminated limit ($\nu \rightarrow -\infty$), $|D(\omega)|$ should approach unity, indicating no diffraction loss.
- The phase of $D(\omega)$ should vary smoothly with frequency and should not exhibit discontinuities or abrupt jumps.

B.2 Double-edge diffraction

For two sequential edges, such as the one-story and two-story houses in the reference geometry:

- The combined diffraction factor $D_1(\omega)D_2(\omega)$ should produce greater attenuation than either edge alone.
- When the edges are close together, the frequency response should show smoother attenuation with mild interference structure.
- When the edges are farther apart, the frequency response should exhibit more pronounced oscillatory structure due to the phase interaction between D_1 and D_2 .
- The combined magnitude should remain physically reasonable and should not exceed unity.

B.3 Ground reflection behavior

For the image-source ground reflection model:

- The direct and reflected paths should interfere constructively or destructively depending on frequency and geometry, producing the expected comb-filter pattern.
- As the receiver height approaches the source height, the spacing between interference minima should increase.
- At very low frequencies, the ground reflection should approach the behavior of a single effective source due to long wavelengths.

B.4 Overall propagation behavior

For the full model (source, ground, and diffraction):

- The total level should decrease monotonically with increasing source–receiver distance when geometry is fixed.
- Introducing an additional edge should never increase the received level.
- The frequency response should remain smooth and physically plausible, with no numerical instabilities or discontinuities.

B.5 Summary of expected monotonicity and signs

Quantity	Expected Behavior	Reason
$\nu(\omega)$	Increases with ω	Fresnel parameter scales as $\sqrt{\omega}$
$ D(\omega) $ (single edge)	Decreases with ν	Higher freq. $\rightarrow$ deeper shadow
$\arg D(\omega)$	Smooth, continuous	Fresnel integrals are continuous
$ D_1 D_2 $	$\leq D_1 $ and $\leq D_2 $	Additional edge cannot increase level
Ground interference spacing	Increases as $h_r \rightarrow h_s$	Path difference decreases
Total level vs. distance	Monotonic decrease	Spherical spreading + losses

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